

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 21-2022: A 17-Year-Old Girl with Fever and Cough

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PRESENTATION OF CASE

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Dr. Samantha D. Martin (Pediatrics): A 17-year-old girl was admitted to this hospital during the coronavirus disease 2019 (Covid-19) pandemic because of fever and cough.

The patient had been well until 10 days before admission, when fever and cough that was productive of clear sputum developed. She also had fatigue, eye redness, sore throat, nasal congestion, rhinorrhea, and myalgias.

Seven days before this admission, the patient was evaluated in the urgent care clinic of another hospital. On examination, there was conjunctival injection in both eyes. The lungs were clear on auscultation, and the remainder of the physical examination was reportedly normal. Testing of a nasopharyngeal specimen for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) RNA was negative. Treatment with acetaminophen and dextromethorphan was recommended.

During the next 2 days, the fever decreased but did not resolve with the use of acetaminophen. New mild chest discomfort occurred with coughing. An expectorated sputum specimen contained flecks of blood, and the patient returned to the urgent care clinic. The physical examination was unchanged, and rapid antigen testing for influenza types A and B was negative. Additional treatment with ibuprofen, benzonatate, and guaifenesin was recommended.

Three days before this admission, the patient had persistent symptoms and was unable to sleep because of the fever and cough. New shortness of breath developed. She was evaluated by a primary care physician at the other hospital in a telehealth visit. Empirical treatment with amoxicillin was started, and close monitoring was recommended. During the next 3 days, the shortness of breath increased in severity, and the patient was instructed to present to the emergency department of this hospital.

On evaluation, the patient reported cough that was worse with exertion and deep breathing, improved with rest, and was associated with pain in the chest and back on the left side. The sputum had streaks of bright red blood. Other symptoms included fatigue, sore throat, nasal congestion, rhinorrhea, myalgias, and pain in

both knees when walking. The eye redness had resolved, but there was mild soreness of the right eye. There was no headache, ear pain, difficulty swallowing, abdominal pain, nausea, vomiting, diarrhea, or rash.

The patient had a history of developmental dysplasia of the hip and was otherwise healthy. She had normal growth and development and had received all routine childhood vaccinations. Medications included amoxicillin, acetaminophen, ibuprofen, benzonatate, guaifenesin, and dextromethorphan. There were no known drug allergies. The patient was born in New England and lived in a suburban area of New England with her mother, father, and brother. She performed well in high school, which she was attending remotely because of the Covid-19 pandemic; she had no known sick contacts. She had traveled to a Caribbean island 1 year earlier. She was a vegetarian and did not drink alcohol, smoke tobacco, or use illicit drugs. Her parents and brother were healthy.

On examination, the temperature was 38.5°C, the heart rate 124 beats per minute, the blood pressure 142/62 mm Hg, the respiratory rate 30 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 35.9. The patient appeared pale, and she coughed and sniffled frequently. She did not use accessory muscles while breathing, and breath sounds in the lower lung fields were decreased. There was no conjunctival injection, lymphadenopathy, joint swelling or tenderness, or rash; the remainder of the physical examination was normal.

The creatinine level was 2.00 mg per deciliter (177 μ mol per liter; reference range, 0.60 to 1.50 mg per deciliter [53 to 133 μ mol per liter]) and the C-reactive protein level 77.5 mg per liter (reference value, <8.0). The hemoglobin level was 6.7 g per deciliter (reference range, 12.0 to 16.0), the hematocrit 21.1% (reference range, 36.0 to 46.0), the white-cell count 11,890 per microliter (reference range, 4500 to 13,000), and the platelet count 526,000 per microliter (reference range, 150,000 to 450,000). Urinalysis revealed 3+ blood and 2+ protein; microscopic examination of the urinary sediment revealed more than 100 red cells per high-power field (reference range, 0 to 2) and 10 to 20 white cells per high-power field

Table 1. Laboratory Data.*

Variable	Reference Range†	On Admission
Hemoglobin (g/dl)	12.0–16.0	6.7
Hematocrit (%)	36.0–46.0	21.1
Platelet count (per μ l)	150,000–450,000	526,000
White-cell count (per μ l)	4500–13,000	11,890
Differential count (per μ l)		
Neutrophils	1800–8100	8810
Lymphocytes	1200–5200	1760
Monocytes	200–1400	880
Eosinophils	0–1000	330
Basophils	0–400	50
Creatinine (mg/dl)	0.60–1.50	2.00
Urea nitrogen (mg/dl)	8–25	22
Sodium (mmol/liter)	135–145	136
Potassium (mmol/liter)	3.4–5.0	4.3
Chloride (mmol/liter)	98–108	99
Carbon dioxide (mmol/liter)	23–32	24
Lactate dehydrogenase (U/liter)	110–210	462
C-reactive protein (mg/liter)	<8.0	77.5
Prothrombin time (sec)	11.5–14.5	15.6
Prothrombin-time international normalized ratio	0.9–1.1	1.3
Activated partial-thromboplastin time (sec)	22.0–36.0	40.7

* To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

(reference value, <10), as well as mixed granular casts and red-cell casts. Nucleic acid testing of a nasopharyngeal specimen for SARS-CoV-2, influenza types A and B, and respiratory syncytial virus was negative. An interferon- γ release assay for *Mycobacterium tuberculosis* was performed, and blood cultures were obtained. Other laboratory test results are shown in Table 1.

Dr. Randheer Shailam: Chest radiography (Fig. 1A and 1B) revealed prominent patchy opacities in the lower lung zones. Computed tomography (CT) of the chest (Fig. 1C through 1F), performed without the administration of intravenous contrast material, revealed peripheral patchy ground-glass opacification with greater involvement of the

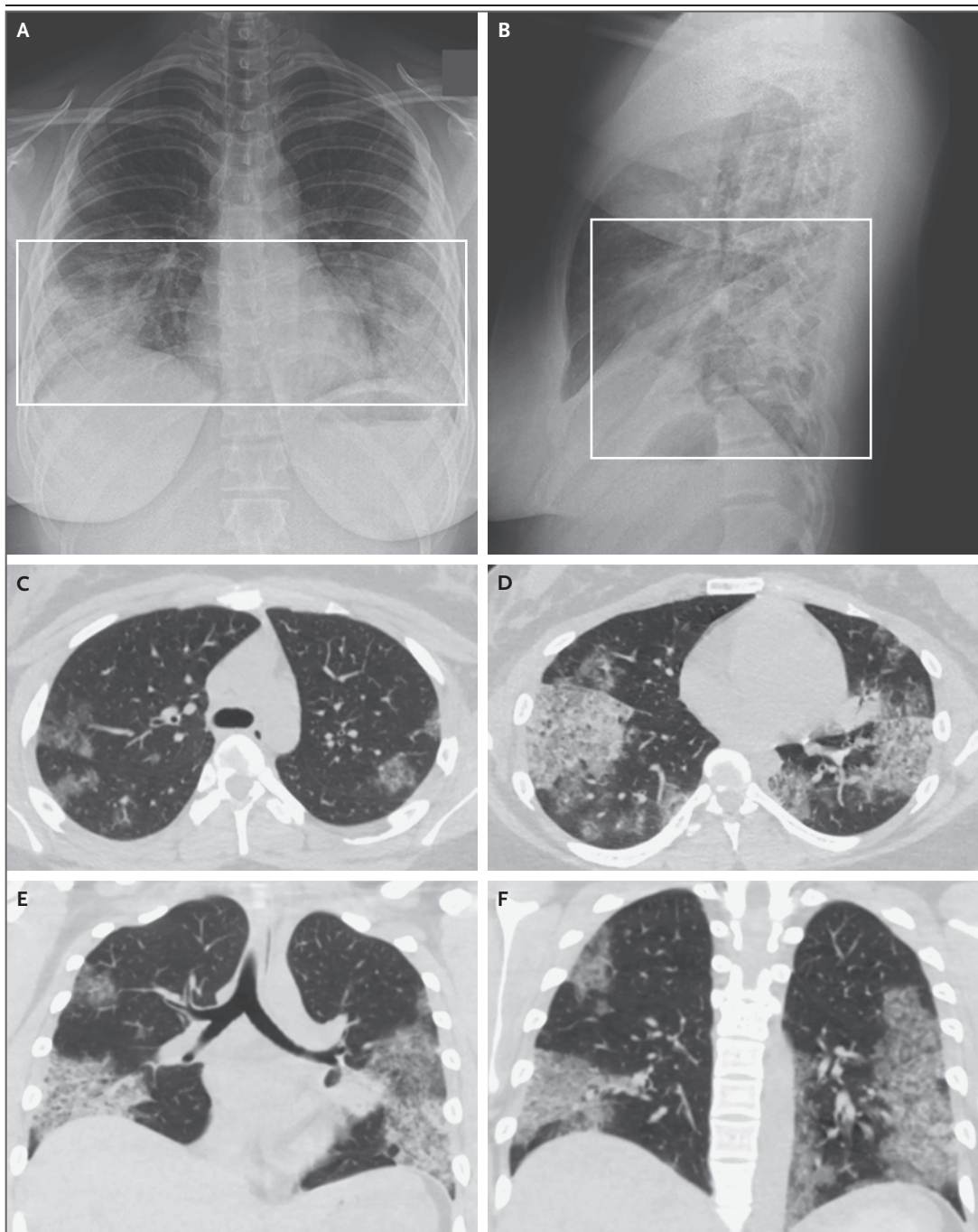


Figure 1. Chest Imaging Studies.

Frontal and lateral radiographs of the chest (Panels A and B, respectively) show patchy airspace opacities, predominantly in the lower lung zones (see boxes). On the lateral radiograph, the opacities are most prominent in the mid-to-posterior aspect of the lungs. There is no evidence of pleural effusion or mediastinal lymphadenopathy. Axial re-formatted CT images of the chest (Panels C and D), obtained without the administration of intravenous contrast material, show peripheral patchy ground-glass opacification with greater involvement of the lower lobes than of the upper lobes, the lingula (left lung), and the middle lobe (right lung). Coronal re-formatted CT images (Panels E and F) show a posterior distribution of opacities.

lower lobes than of the upper lobes, the lingula (left lung), and the middle lobe (right lung). There were also areas of normal-appearing lung. These findings are not specific and can be seen with infectious causes such as bacterial pneumonia and Covid-19, as well as with noninfectious causes such as aspiration, pulmonary hemorrhage, and less likely, alveolar proteinosis.

Dr. Martin: Ceftriaxone, azithromycin, and intravenous fluids were administered. A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Mary C. Ottolini: This previously healthy teenage girl with obesity presented to the emergency department during the Covid-19 pandemic with prolonged fever, cough, fatigue, and other symptoms of upper respiratory infection. Conjunctivitis had occurred and resolved in both eyes, with residual pain in the right eye. Pleuritic pain and dyspnea had developed. The cough, which had initially been productive of clear sputum, had progressed to hemoptysis.

On the basis of the patient's history and physical examination, the predominant problems that warranted hospitalization were the prolonged unexplained fever with progression to a systemic inflammatory response syndrome and the hemoptysis. In developing a differential diagnosis, I will first consider causes of prolonged fever, including infections and autoimmune diseases that are associated with a systemic inflammatory response. Although a 10-day history of fever is unusual for a typical viral process, it does not meet criteria to be classified as a fever of unknown origin.^{1,2}

Other features of the patient's presentation that will ultimately need to be explained are hemoptysis and bilateral conjunctivitis with residual eye pain. Hemoptysis can be due to capillaritis from necrotizing pneumonia; it can also be a component of systemic vasculitis.^{3,4} Self-resolving bilateral conjunctivitis with residual eye pain is an unusual finding that could reflect anterior eye inflammation as a manifestation of systemic inflammatory disease.⁵

In addition, several notable laboratory findings need to be considered, including severe anemia with a normal white-cell count, an increased platelet count, and mildly elevated results of coagula-

tion tests. The red-cell indexes and a description of the morphologic features of the red cells are not provided; these could be helpful in the evaluation of anemia.

CAUSES OF ANEMIA AND OTHER FINDINGS

The most likely causes of anemia in a previously healthy adolescent girl are bone marrow suppression due to prolonged illness, hemolysis due to oxidative stress in the presence of glucose-6-phosphate dehydrogenase (G6PD) deficiency, or iron-deficiency anemia due to previously undiagnosed menorrhagia. Blood loss from pulmonary hemorrhage is also possible but does not seem likely initially, given the description of the hemoptysis as "flecks" and "streaks" of blood rather than substantial bleeding.

Urinalysis showed an active sediment with clinically significant amounts of blood, protein, and red-cell casts. Blood testing showed evidence of kidney injury, including an elevated creatinine level. A ratio of urea nitrogen to creatinine of more than 10 and the presence of blood, protein, and red-cell casts in urine are features consistent with a diagnosis of glomerulonephritis. An elevated lactate dehydrogenase level could be due to hemolysis, vasculitis, or muscle or liver injury. A C-reactive protein level almost 10 times the upper limit of the normal range is indicative of severe systemic inflammation. Patchy opacities in the lower lobes could be due to pneumonia, pulmonary edema, or pulmonary hemorrhage.

ADENOVIRUS INFECTION AND BACTERIAL PNEUMONIA

Infection with adenovirus, especially adenovirus serotype 14, begins with symptoms of upper respiratory infection, arthralgias, and myalgias and can cause clinically significant inflammation of the bulbar and palpebral conjunctiva. Adenovirus serotype 14 infection has been associated with fatal necrotizing pneumonia, alveolar hemorrhage, and hemoptysis.⁶ In this patient, the bilateral patchy opacities are consistent with diffuse pulmonary inflammation. The severe anemia and coagulation abnormalities are not consistent with adenovirus infection, and acute glomerulonephritis due to adenovirus infection is extremely rare.⁷

Adenovirus and other viral infections can be complicated by secondary bacterial pneumonia,

which is often due to *Staphylococcus aureus* or *Streptococcus pyogenes*. Secondary bacterial pneumonia can cause prolonged fever and hemoptysis, especially if it is complicated by empyema or abscess. In this patient, the absence of focal consolidation or effusion on chest imaging makes bacterial pneumonia unlikely. The anemia was more severe than would be expected with bone marrow suppression due to infection, although such severe anemia could be due to hemolysis in the presence of G6PD deficiency. The coagulation abnormalities could be due to disseminated intravascular coagulation or sepsis, but these conditions are associated with a more ill appearance and an elevated white-cell count with a left shift. Postinfectious glomerulonephritis can occur after infection with certain nephritogenic group A streptococci, but the onset would be a bit early if the bacterial infection occurred several days into the viral infection.⁸

COVID-19 AND MULTISYSTEM INFLAMMATORY SYNDROME IN CHILDREN

This patient had some risk factors for severe complications of Covid-19, including her age (i.e., the fact that she was not a young child) and obesity.^{9,10} The respiratory issues, including hemoptysis, and the systemic symptoms — prolonged fever, arthralgias, myalgias, and fatigue — are classic features of Covid-19. The findings on chest radiography are also consistent with Covid-19, but if the diffuse pneumonitis were due to Covid-19, hypoxia would be expected by the time of her presentation to the emergency department.^{11,12} Severe anemia and glomerulonephritis are not common findings seen with acute Covid-19, and persistent unilateral eye pain is also unusual.

Multisystem inflammatory syndrome in children (MIS-C) due to Covid-19 is a possible diagnosis in this case, given the multiorgan systemic involvement and evidence of severe inflammation. The patient met some of the established criteria for MIS-C, including an age younger than 21 years, a temperature higher than 38°C for at least 24 hours, and laboratory evidence of inflammation.¹³ She had a severe illness with involvement of the renal, respiratory, and hematologic systems that led to hospitalization. Involvement of the cardiac and gastrointestinal systems is common with MIS-C, whereas the type of renal and hematologic involvement de-

scribed in this patient is atypical.¹⁴ Renal injury due to acute tubular necrosis from inadequate perfusion is more common than acute glomerulonephritis.¹⁵ The onset of fever is earlier than would be expected with MIS-C, which typically develops a minimum of 2 weeks after the onset of Covid-19 symptoms.^{16,17} Furthermore, this patient had multiple negative tests for Covid-19 and had no known exposure, since she was attending school remotely.

SYSTEMIC LUPUS ERYTHEMATOSUS

This patient had many findings consistent with systemic lupus erythematosus (SLE), including severe inflammation, prolonged fever, arthralgias, fatigue, glomerulonephritis, and coagulopathy.¹⁸ The pulmonary opacities, hemoptysis, and anemia are also consistent with this diagnosis, although pleural effusions are more common than pulmonary opacities alone.¹⁹ SLE often occurs in adolescent girls, but it is usually associated with an insidious onset of symptoms, whereas this patient had a relatively acute symptom onset. Also, the conjunctivitis with residual eye pain is an uncommon manifestation. To establish the diagnosis of SLE on the basis of the 2012 Systemic Lupus Erythematosus International Collaborating Clinics criteria, a patient must meet at least four of the clinical criteria and have associated laboratory findings, unless there is a histologic diagnosis of lupus nephritis.²⁰ This patient met only three of the clinical criteria — hemolytic anemia, kidney disease, and joint (knee) inflammation — which makes the diagnosis of SLE unlikely. However, it is important to perform a laboratory evaluation to determine whether autoantibodies associated with SLE are present.

GRANULOMATOSIS WITH POLYANGIITIS

This patient met the 2008 criteria for the diagnosis of granulomatosis with polyangiitis (GPA) from the European Alliance of Associations for Rheumatology, Pediatric Rheumatology International Trials Organization, and Pediatric Rheumatology European Society.²¹ To establish the diagnosis of GPA, a patient must meet at least three of the following six criteria: lung involvement,²² kidney involvement,²³ upper airway involvement, inflammation of the laryngotracheobronchial tract, a positive test for antineutrophil cytoplasmic antibody (ANCA), and granulomatous inflammation located in an arterial wall or

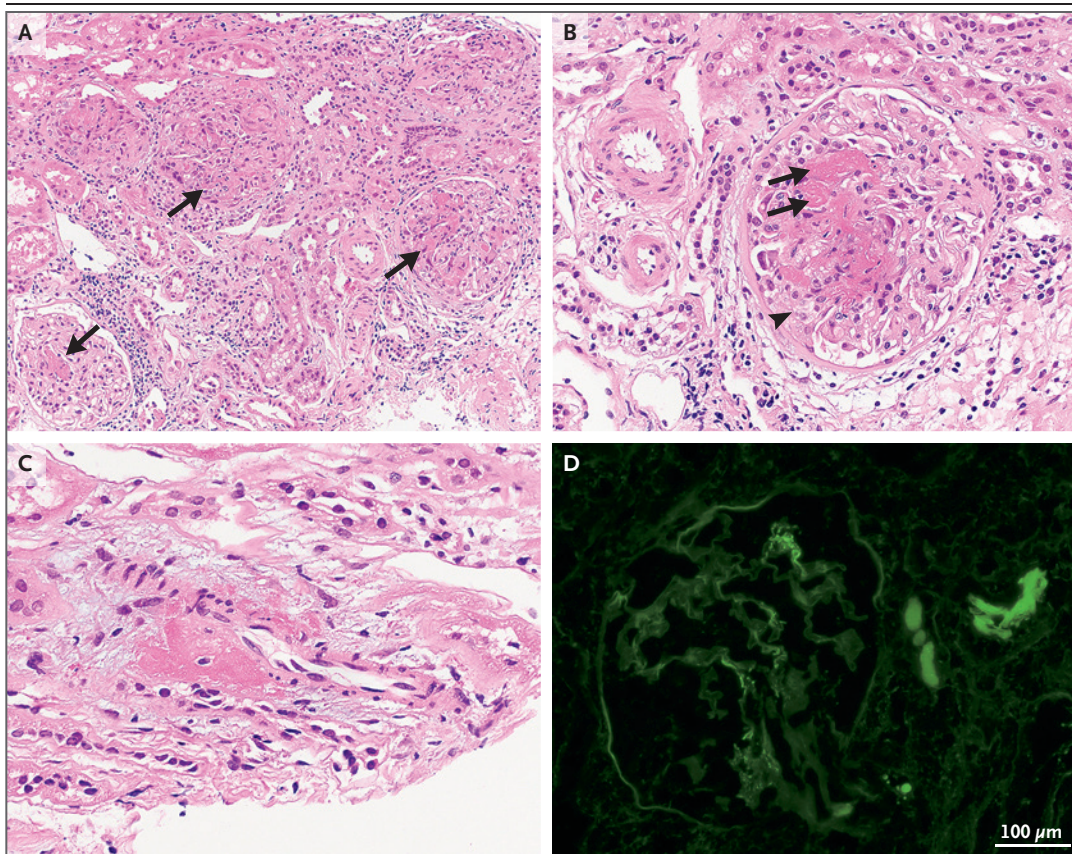


Figure 2. Kidney-Biopsy Specimen.

On hematoxylin and eosin staining, the renal cortex shows many abnormal glomeruli with segmental to global necrotizing lesions (Panel A, arrows). At higher magnification, a glomerulus shows fibrin (Panel B, arrows) associated with nuclear fragmentation, as well as a cellular crescent (Panel B, arrowhead). The renal medulla shows an arteriole with wall necrosis (Panel C). A direct immunofluorescence assay for IgG shows trace segmental mesangial staining (Panel D).

in the perivascular or extravascular space. This patient had involvement of the lungs (bilateral patchy infiltrates and hemoptysis), kidneys (glomerulonephritis), and upper airway (prolonged rhinorrhea and nasal congestion). In addition, her systemic inflammatory response with fever and coagulopathy and her severe anemia that was determined to be due to pulmonary hemorrhage are consistent with a diagnosis of GPA. It would be helpful to know more about her “conjunctivitis” to determine whether it was due to inflammation of the anterior chamber or an infectious process. The acute onset of GPA in a previously healthy girl is a bit unusual, but there have been case reports of acute infections that uncover an evolving ANCA-associated vasculitis.²⁴ Although additional laboratory tests would

help to refine the differential diagnosis, the definitive test in this case would be a kidney biopsy to establish the cause of the kidney disease.

DR. MARY C. OTTOLINI'S DIAGNOSIS

Granulomatosis with polyangiitis.

PATHOLOGICAL DISCUSSION

Dr. Ivy A. Rosales: The diagnostic procedure in this case was a core-needle biopsy of the kidney. On light microscopy of biopsy specimens (Fig. 2A, 2B, and 2C), the cortex showed 17 glomeruli, of which 15 (88%) had cellular crescents. Crescents occur after rupture of the glomerular capillary wall and are associated with proliferation of

epithelial cells and infiltration of inflammatory cells in Bowman's space. Some glomeruli had necrotizing lesions, which were characterized by fibrin, cellular breakdown, and destruction of the glomerular capillaries. Unaffected glomeruli had normal capillary walls and did not have hypercellularity. Tubular injury and many red-cell casts were present. The medulla showed interstitial inflammation composed of lymphocytes, eosinophils, and neutrophils. An arteriole had wall inflammation and necrosis. In summary, these findings are indicative of necrotizing and crescentic glomerulonephritis with medullary small-vessel vasculitis.

A direct immunofluorescence assay of biopsy specimens for IgG (Fig. 2D), IgA, IgM, C3, C1q, and light chains revealed nonspecific, trace, segmental staining along the glomerular capillary walls and mesangium, as well as bright, intense staining of fibrin along glomerular crescents. The tubular basement membranes and vessels were negative for all reactants. These findings are indicative of pauci-immune glomerulonephritis. There was a lack of glomeruli in the sample, so electron microscopy was not performed.

An indirect immunofluorescence assay performed on fixed neutrophils from patient serum showed ANCA with a cytoplasmic pattern of staining (c-ANCA). An enzyme-linked immunosorbent assay showed antibodies to proteinase 3 at levels of up to 21,504 units. Taken together, these findings are diagnostic of ANCA-associated necrotizing and crescentic glomerulonephritis.

In patients with ANCA-associated vasculitis, the pathogenesis of acute injury is mediated by neutrophils and monocytes and is associated with multiple factors, including genetics, loss of T-cell regulation, environmental factors, and variation in response to injury by T cells and macrophages.²⁵ The mechanism through which ANCA causes vasculitis is driven by the activation of neutrophils. Activated neutrophils release factors that then activate the alternative pathway and generate the neutrophil chemoattractant C5a.²⁵ This results in primed and activated neutrophils, which attach to and penetrate the endothelium and vessel walls, leading to injury and necrosis. ANCA-associated vasculitis may resolve or result in scarring of the affected organ.²⁵

Although many ANCA antigens have been identified, the major ones are proteinase 3 and myeloperoxidase.²⁶ ANCA-associated vasculitis

can involve medium and small vessels, and more specific diagnoses can be made within each type, depending on organ involvement.²⁷ Overall, the clinical and pathological features of this case are consistent with a diagnosis of GPA.

PATHOLOGICAL DIAGNOSIS

Antineutrophil cytoplasmic antibody–associated necrotizing and crescentic glomerulonephritis with focal medullary arteriolitis and capillaritis.

DISCUSSION OF MANAGEMENT

Dr. Amita Sharma: Once the diagnosis of GPA is established, it is critical to initiate therapy as soon as possible. Without treatment, the crescentic glomerulonephritis in this patient would be likely to progress to severe kidney impairment, placing her at risk for fatal alveolar hemorrhage.

Strategies to treat patients with GPA are predicated on clearing B cells and plasma cells that produce ANCA, removing the antibody, and blocking the effect of the antibody on neutrophils, thus mitigating tissue damage. Pulse glucocorticoid therapy decreases levels of circulating ANCA-producing B cells, plasma cells, and plasma blasts, and continuously administered glucocorticoid therapy blocks priming of neutrophils, thus decreasing overall inflammation.

This patient was treated with induction therapy that consisted of high-dose glucocorticoids in combination with cyclophosphamide and rituximab. Cyclophosphamide targets ANCA-producing plasma cells, and rituximab targets ANCA-producing B cells. Combination therapy with cyclophosphamide, rituximab, and glucocorticoids is associated with a remission rate of 88%, as compared with a rate of 60% with rituximab and glucocorticoids. In addition, combination therapy allows for a shorter cyclophosphamide course, thus decreasing the risk of cancer and infertility, and allows for quick tapering of the glucocorticoid course, thus decreasing the risk of acne, obesity, and osteoporosis.^{28,29} Despite these interventions, the patient's serum creatinine level increased to 4.38 mg per deciliter (387 μ mol per liter), and the hemoglobin level decreased to 6.0 g per deciliter. Her respiratory status was stable, but she began to receive supplemental oxygen during sleep because of occasional desaturation.

Plasma exchange was considered in this pa-

tient, but it was not performed because her respiratory status was relatively stable while she was breathing ambient air and the PEXIVAS study did not show a benefit.³⁰ Rather than opting for plasma exchange, we administered eculizumab therapy. Eculizumab targets C5, blocking complement activation and thus interrupting the microvascular necroinflammation loop, which is driven by neutrophil activation and neutrophil extracellular traps.³¹ After the patient received the first dose of eculizumab, the gross hematuria resolved within 48 hours, the serum creatinine level decreased, and the C3 level increased. Although patients with GPA may not have overtly low serum complement levels, a study showed that when patients were stratified according to high or low levels of C3 within the normal range, those with low levels had worse outcomes. Patients with low C3 levels (<120 mg per deciliter) had higher serum creatinine levels and an unfavorable odds ratio of 3.5 for end-stage kidney disease.³² Before the administration of eculizumab, this patient had a C3 level of 116 mg per deciliter (reference range, 81 to 157) and a C4 level of 21 mg per deciliter (reference range, 12 to 39). After the administration of eculizumab, the patient's C3 level increased to 158 mg per deciliter and the C4 level to 48 mg per deciliter.

The patient received eculizumab therapy for 3 months without any clinically significant side

effects. She also received meningococcal vaccine and prophylactic oral penicillin. Cyclophosphamide therapy was discontinued after 8 weeks, glucocorticoid therapy was tapered and then discontinued within 6 months, and rituximab was discontinued after 6 months.

At the patient's most recent follow-up visit, 14 months after the diagnosis of GPA, she had a serum creatinine level of 1.10 mg per deciliter (97 μ mol per liter) with resolving proteinuria and an ANCA titer of less than 1%. We are currently in the process of obtaining prior authorization for avacopan, an orally administered C5a receptor inhibitor that requires prior authorization from the patient's insurance company. The patient is being monitored closely for the need of repeat rituximab infusion. Her IgG level decreased substantially after treatment with rituximab, so subcutaneous IgG replacement therapy has been initiated.

FINAL DIAGNOSIS

Granulomatosis with polyangiitis.

This case was presented at the Tufts University School of Medicine postgraduate course "New England Pediatric Hospital Medicine 2022," directed by Dr. Chadi El Saleeby and Dr. Kerstin Zanger.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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